

Application Flow and Implementation

The runtime flow begins in `run_app.py`, where the Flask app is configured and the SQLite database path is registered. The system then loads three blueprints for portal, staff, and admin functions.

Operational Flow

- On startup, the application ensures that the default manager account exists.
- New guides can register, but they cannot log in until a manager verifies them.
- Managers can create expeditions and optionally assign a verified guide during creation.
- Trekkers can search only open expeditions and reserve seats when availability is greater than zero.
- Guides can update status to Open, Closed, or Completed and can adjust available seats within maximum capacity.

Seed Data Support

The seed script creates one manager, one guide, one trekker, four expeditions, and one sample booking. This makes the system immediately demo-ready and also helps validate relationships between users, expeditions, and bookings.

Implementation Highlights

Concern	Current Approach
Password security	Passwords are hashed using PBKDF2-SHA256 through Werkzeug helpers.
Role protection	Session-based role checks guard dashboard and action routes.
Account control	Verification and suspension rely on Boolean status fields in the users table.
Seat management	Expedition availability is reduced on booking and can be adjusted by the assigned guide.